

# YINGDAN SHI

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## EDUCATION

### Illinois Institute of Technology

Ph.D. in Computer Engineering. Advisor: [Prof. Ren Wang](#)

Aug. 2024 - Present

GPA: 4.0/4.0

### Soochow University

M.E. in Computer Technology. Advisor: Prof. Jingya Zhou

Sep. 2021 - Jun. 2024

GPA: 3.7/4.0

### Hangzhou Normal University

B.E. in Computer Science and Technology.

Sep. 2017 - Jun. 2021

GPA: 3.8/4.0

## RESEARCH INTERESTS

- Post-training, Agentic System, LLMs, MLLMs
- AI Safety, Alignment, Trustworthy Algorithms, Machine Unlearning, Continual Learning, Model Editing

## PUBLICATIONS

(\* means Equal Contribution)

- [1] [Yingdan Shi](#), Ren Wang. “Exploring Nonlinear Pathway in Parameter Space for Machine Unlearning.” **ICML**, 2026. [[Paper](#)] [[Code](#)]
- [2] [Yingdan Shi](#), Sijia Liu, Kaize Ding, Ren Wang. “Tackling Fake Forgetting through Uncertainty Quantification.” **ICML**, 2026. [[Paper](#)] [[Code](#)]
- [3] Jane Downer\*, [Yingdan Shi](#)\*, Ziyang Liu\*, Ren Wang, Binghui Wang. “Watermarking Graph Neural Networks via Explanations for Ownership Protection.” **ICML**, 2026. [[Paper](#)] [[Code](#)]
- [4] [Yingdan Shi](#), Xiang Xu, Kaize Ding, Alfred O. Hero, Ren Wang. “When Models Fail to Forget: Plasticity Collapse in Continual Unlearning.” In submission, 2026.
- [5] [Yingdan Shi](#), Jingya Zhou, Congcong Zhang. “Towards Node-Level Influence Estimation in Social Networks.” In submission, 2025.
- [6] [Yingdan Shi](#), Jingya Zhou, Congcong Zhang, Zhenyu Hu. “Influence contribution ratio estimation in social networks.” *Information Sciences [J]*, 2025. [[Paper](#)]
- [7] [Yingdan Shi](#), Jingya Zhou, Congcong Zhang. “DySuse: Susceptibility Estimation in Dynamic Social Networks.” *Expert Systems with Applications [J]*, 2023. [[Paper](#)]
- [8] Zhaoying Wang, [Yingdan Shi](#), Xiang Liu, Can Chen, Jun Wen, Ren Wang. “HODDI: A Dataset of High-Order Drug-Drug Interactions for Computational Pharmacovigilance.” In submission, 2025.
- [9] Zhenyu Hu, Jingya Zhou, Wenqi Wei, Congcong Zhang, [Yingdan Shi](#). “Predicting Cross-domain Collaboration Using Multi-task Learning.” *Expert Systems with Applications [J]*, 2024. [[Paper](#)]
- [10] Zhenyu Hu, Jingya Zhou, Congcong Zhang, [Yingdan Shi](#). “Explainable Cross-Domain Collaborator Recommendation.” *CSCWD 2024*. [[Paper](#)]
- [11] Congcong Zhang, Jingya Zhou, Jin Wang, Jianxi Fan, [Yingdan Shi](#). “Fairness-Aware Competitive Bidding Influence Maximization in Social Networks.” *IEEE Transactions on Computational Social Systems [J]*, 2024. [[Paper](#)]
- [12] Congcong Zhang, Jingya Zhou, Wenqi Wei, [Yingdan Shi](#). “Order-sensitive Competitive Revenue Maximization for Viral Marketing in Social Networks”. *Information Sciences [J]*, 2025. [[Paper](#)]

## RESEARCH EXPERIENCE

### **TIML**, Illinois Institute of Technology

Research Assistant. Advisor: [Prof. Ren Wang](#)

Chicago, IL

Aug. 2024 - Present

- **Plasticity Collapse in Continual Unlearning.** Identified a novel phenomenon termed *Unlearning Plasticity Collapse* in continual unlearning, manifesting as two failure modes: forward failure, where the model progressively loses its ability to forget new data, and backward failure, where previously forgotten data is re-memorized. Through theoretical analysis of continual unlearning dynamics, we demonstrate that successive unlearning operations accumulate geometric constraints in parameter space. Empirical results across image classification, image generation (DDPM), and multimodal tasks (LLaVA) consistently validate this finding.
- **LLM Knowledge Entanglement.** Rather than representing knowledge as a single vector, proposed characterizing knowledge via the subspace of hidden states and systematically investigating inter-knowledge entanglement. Leveraging this entanglement structure provides principled guidance for knowledge editing and unlearning in large language models.
- **Mode Connectivity Unlearning.** Revealed that existing linear unlearning methods suffer from a weight entanglement issue that prevents ideal forgetting. Exploited mode connectivity to explore an unlearning pathway from a nonlinear perspective, enabling plug-and-play integration with existing unlearning methods and mitigating both over-forgetting and under-forgetting.
- **Fake Forgetting.** Observed that existing machine unlearning methods overlook prediction uncertainty. Through conformal prediction, we show that data points diagnosed as forgotten based on unlearning accuracy can still be exposed, revealing a critical gap in current evaluation. Proposed a novel uncertainty-aware metric and framework to address this limitation.
- **GNN Watermarking.** Identified that the *de facto* backdoor-based watermarking approach relies on training data manipulation, introducing ownership ambiguity through misclassification and vulnerability to data poisoning attacks that can disrupt the backdoor mechanism. Proposed an explanation-based watermarking scheme that preserves the advantages of backdoor-based methods without data manipulation, thereby eliminating ownership ambiguity and data dependencies.

## WORK EXPERIENCE

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|---------------------------------------------------------------|------------------------------------------------|
| <b>Illinois Institute of Technology</b><br>Research Assistant | <i>Chicago, IL</i><br>Aug. 2024 - Present      |
| <b>Meituan Inc.</b><br>Business Analyst Internship            | <i>Beijing, China</i><br>May. 2024 - Jul. 2024 |
| <b>Soochow University</b><br>Research Assistant               | <i>Suzhou, China</i><br>Aug. 2021 - Jun. 2024  |

## AWARDS AND HONORS

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Excellent Master's Thesis Award (2024)  
Outstanding Master's Award (2024)  
National Scholarship (2023)  
Outstanding Undergraduate Award (2021)  
First Prize Scholarship (2020)

## ACADEMIC SERVICE

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**Conference Reviewer:** ICML 2026, SIGIR 2026, NeurIPS 2026  
**Journal Reviewer:** World Wide Web, Scientific Reports, Dynamic Games and Applications, Humanities and Social Sciences Communications, Knowledge and Information Systems, Social Network Analysis and Mining

## SKILLS

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**Language:** Mandarin (Native), English (Professional)  
**Programming:** Python, SQL  
**Libraries / Softwares:** PyTorch, HuggingFace, Scikit-Learn, Numpy  
**Specialized Tools:** Geometric (PyG), NetworkX, Git, Linux/Bash, LaTeX  
**Developer Tools** Git, Docker, Vim, VSCode, Cursor, Claude Code, PyCharm